

PHANIDHAR AKULA

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Simulation Systems, Digital Twins & Geospatial Computing • Authorized to work in the U.S. Oct 2026 – Oct 2029 (OPT + STEM OPT)

EDUCATION

Miami University | M.S. in Computer Science

Aug 2024 – Aug 2026

Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Assistantship

Thesis: SimForge – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (committee of three faculty advisors)

Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science

Aug 2020 – May 2024

Hyderabad, India • CGPA: 8.2/10 • Two peer-reviewed publications (CNN soil classification, LDA topic modeling)

SELECTED PROJECTS

SimForge | Master's Thesis – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation

Aug 2024 – Present

Python, OpenMP, multiprocessing, GPU-accelerated simulation, SLURM, OSC HPC (Pitzer / Cardinal), SUMO, MATSim, DTALite, OSM, US Census PUMS

- ▶ Integrated three heterogeneous traffic engines (SUMO micro+meso, MATSim queue-mobsim, DTALite DTA) under one canonical five-file scenario bundle (network, demand, signals, config, manifest with SHA-256 hashes); byte-identical reproducibility guarantees across re-runs with code-enforced fair-comparison contract.
- ▶ Designed a state-aware BFS heuristic pre-router with OSM turn-restriction enforcement and strongly connected component (SCC) feasibility filtering, forcing engines onto byte-identical canonical routes so any cross-engine travel-time delta is provably an engine-internal property, not an input asymmetry.
- ▶ Compressed BFS pre-routing bottleneck from ~68h to ~5–8h (~25–30× speedup) on NYC 500K-trip scenarios via canonical-route deduplication, 16-way multiprocessing.Pool parallelism, and a JSONL content-addressed cache; deployed across two OSC HPC clusters with cluster-specific SBATCH tuning.
- ▶ Scaled to ~80K nodes / ~200K directed links across Chicago, NYC, LA at five demand tiers (1K → 500K trips), calibrated against US Census PUMS microdata (2.4M–6.8M persons) via ModelGen, lifting realism from ~20–40% baseline to ~70–72%. Validated by ~570 pytest tests, mutation tests on load-bearing modules, byte-identity guards, and Student's-t 95% CIs; 76% coverage.
- ▶ Evaluated GPU-accelerated traffic simulators (LPSim and QarSUMO on NVIDIA V100) and additional engines (POLARIS) during the research phase; scoped the final framework to three CPU-based engines after analyzing GPU performance and accessibility limitations.

Cityscape | Open-Source Population Synthesis & Geospatial Demand Generation

Aug 2025 – Present

C++, OpenMP, OSM / Geofabrik, US Census PUMS, OSC HPC (Pitzer)

- ▶ Authored three merged C++ pull requests (github.com/raodj/cityscape, PRs #1–#3) to an open-source PUMS-driven population synthesizer and trip-demand generator. Fixed OSM building-classification and orphan-ring logic across five metros, collapsing synthetic-home ratios (e.g. LA ~70% → ~5%) and removing ~1.25M stranded-population artifacts, affecting well over 1M building records.
- ▶ Exposed search-radius CLI flags and ran a 4-city, 16-combination HPC parameter sweep on OSC Pitzer, raising worst-case calibration R² from 0.812 to 0.917. Contributions integrated into the SimForge thesis demand pipeline.

LumiAI | Production AI Education Platform – studywithlumi.com

Nov 2025 – Present

React, TypeScript, Supabase (PostgreSQL/Auth/Storage), LLMs, Vercel serverless, custom domain

- ▶ Solo-architected, built, deployed, and operate a production AI tutoring platform serving 55+ active student users; document-grounded AI chat, auto-generated quizzes and flashcards, SM-2 spaced-repetition review, and voice mode. Owned the full stack with row-level security, server-side API-key isolation, Google OAuth, and an admin analytics panel.

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE

Aug 2025 – May 2026

Oxford, OH • Research group of Dr. DJ Rao

- ▶ Graduate research in the simulation / digital-twin group; lead developer on Cityscape (above), with work integrated into the SimForge thesis pipeline. Co-administer the Grand Challenges Scholars Program (GCSP) mentoring undergraduate research.
- ▶ Prior: SWE intern (Smart Pathfinders, 2024) and freelance full-stack engineer for three startup clients (2023–2024).

LEADERSHIP

- ▶ President, Graduate Students of Color Association (104 members; Certificate of Appreciation, 2026) and elected Board Member, Graduate Student Council, Miami University (Aug 2025 – May 2026) – led recruitment, events, cross-team coordination, and reviewing student petitions.

TECHNICAL SKILLS

Languages: Python (advanced), C++, JavaScript, TypeScript, SQL, Java (basic)

AI/ML & Data: PyTorch, TensorFlow, CNNs, LDA / topic modeling, LLMs, REST APIs, GraphQL, MySQL, PostgreSQL, AWS

HPC & Systems: OpenMP, multiprocessing, GPU-accelerated simulation, SLURM, OSC clusters (Pitzer / Cardinal), Linux, Docker, Git

Engineering & Evaluation: benchmarking & evaluation frameworks, statistical validation (95% CIs), byte-identity reproducibility guards, pytest, mutation testing, code review

Simulation & Geospatial: SUMO, MATSim, DTALite, agent-based & traffic simulation, digital twins, OSM / Geofabrik, US Census PUMS pipelines, network modeling

PUBLICATIONS & RECOGNITION

▶ Akula, P., Stojanovski, S., & Rao, D. M. "CITYSCAPE: A Digital Twin Framework for Temporospatial Modeling of Work-Commute Dynamics and Roadway Infrastructure Demand" *Frontiers* (first author; under review, 2026).

▶ Two peer-reviewed publications in IJSREM (2023): "Topic Modelling of Web Pages with LDA Methods" (DOI: 10.55041/IJSREM27350) · "Soil Image Classification using Deep Learning" (DOI: 10.55041/IJSREM23138).